

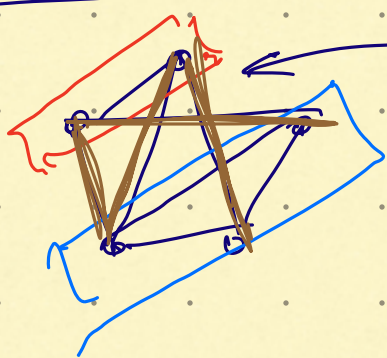
(a) Different representation
(New algorithms)

↳ Backend $\leftarrow$ QAC

(b) A cool feature (Multiple solutions)
New technique

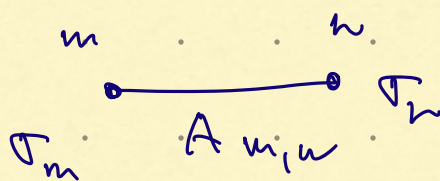
(c) Re-implementation of known algorithms.
(Obviously sequential) $\rightarrow$ Parallelizable

Max-cut problem



cut edges

Their weight is our metric



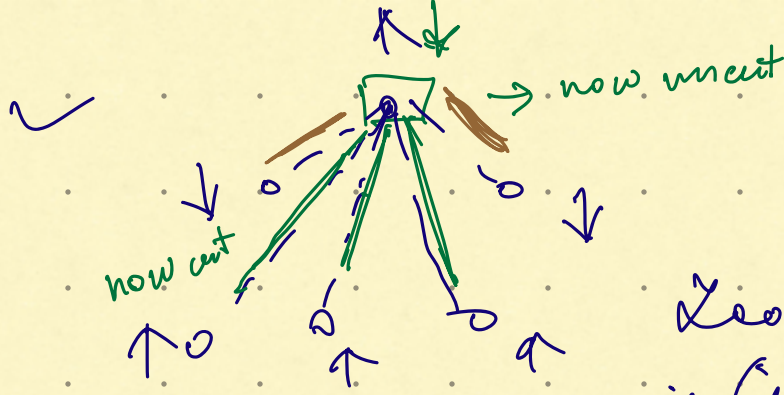
$$x_{mn} = \frac{1}{2} (1 - \sigma_m \sigma_n)$$

$$\underline{C(\underline{\sigma})} = \frac{1}{2} \sum_{mn} A_{mn} \frac{1}{2} (1 - \sigma_m \sigma_n)$$

$$= \frac{1}{2} \sum_{m,n} A_{m,n} - \frac{1}{4} \sum \underline{A_{m,n} \sigma_m \sigma_n}$$

$$= \underline{\frac{1}{2} W - \frac{1}{2} H(\underline{\sigma})}$$

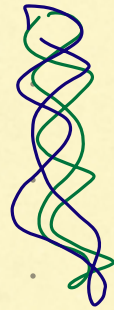
Local search $\approx$ greedy search



$$\sigma_0$$

$$C(\sigma_0)$$

Look at n & σ_n
if (weight cut edges
incident to σ_n)



< (1-1-1 uncut
edges incident to n)

then $\sigma_n \rightarrow -\sigma_n$

after this (weight-cut) > (weight-uncut)

$$\Rightarrow \underline{C(\sigma') > C(\sigma)}$$

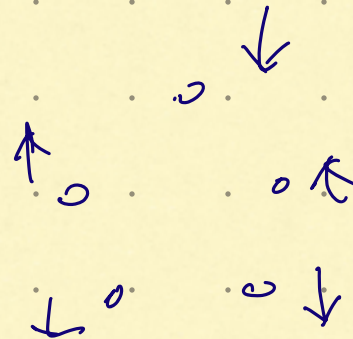
now

$$-\infty < C(\sigma_0) \dots C_{\max} < \infty$$

This algo stops in finite # of steps.

1-opt LS.

Final distribution



Simple flip will not increase cut.

$$\left[F_m = \sum_n A_{mn} \underline{\underline{\sigma_m \sigma_n}} \right] \quad \begin{array}{c} m \\ \swarrow \quad \searrow \\ \sigma \quad \sigma \\ \nwarrow \quad \nearrow \\ n \end{array}$$

if $\sigma_m \sigma_n = -1$

$\Rightarrow (m, n)$ is cut

$\dots = 1$

$\Rightarrow \underline{(m, n) \text{ is uncut.}}$

$$F_m = W_m(\text{cut}) - W_m(\text{uncut}),$$

if $F_m < 0$ we flip σ_m ($\sigma_m \rightarrow -\sigma_m$)

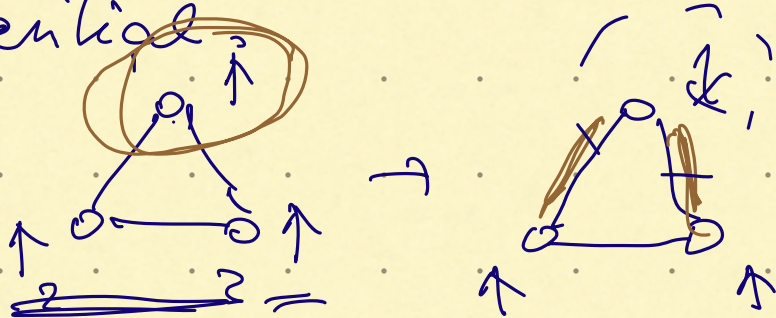
$F_m > 0$ keep the same

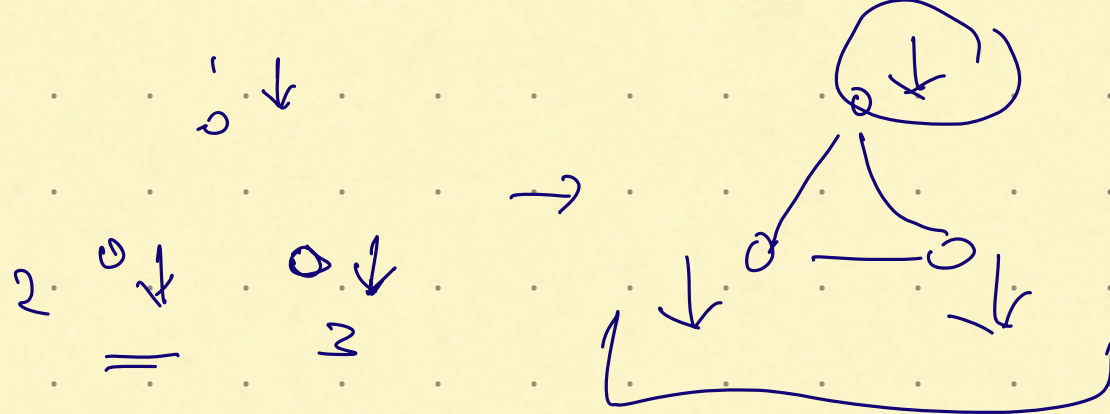
$F_m = 0$ (?) $\left. \begin{array}{l} \sigma_m \text{ doesn't} \\ \text{matter} \end{array} \right\}$

keep the same.

$\mathcal{Q}$ is stable if none of the spins flips.

IS is sequential





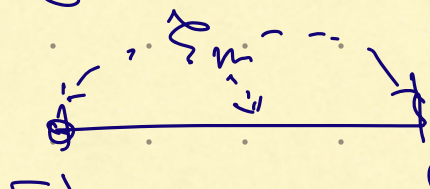
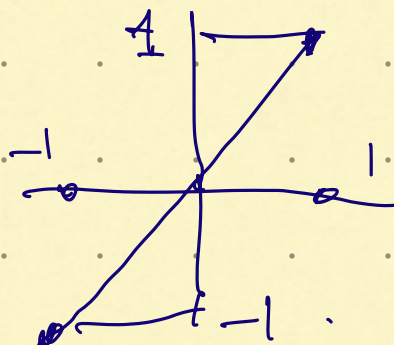
It's tough to parallel.

Not R-spin

$$C(\sigma) = \frac{1}{4} \sum A_{m,n} (1 - \sigma_m \sigma_n)$$

$$A_{m,n} \sim 0$$

$$\sigma_m \rightarrow \xi_m \in [-1, 1]$$



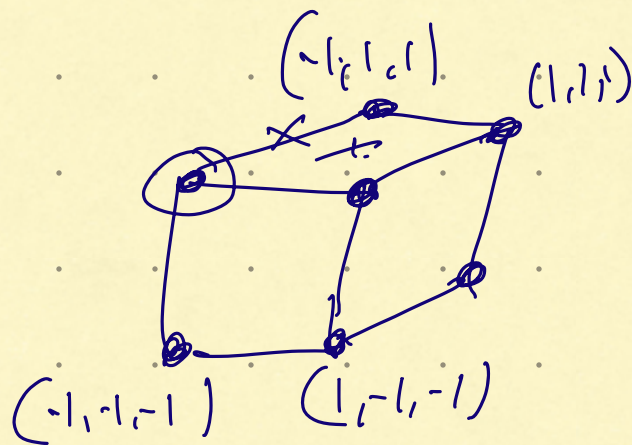
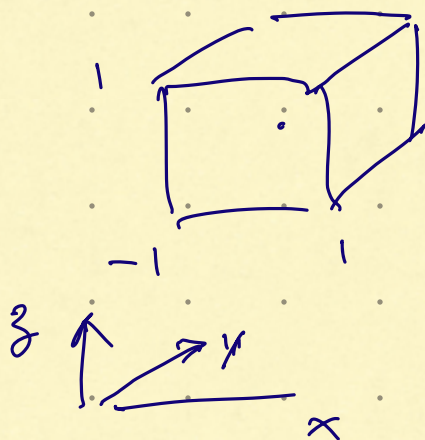
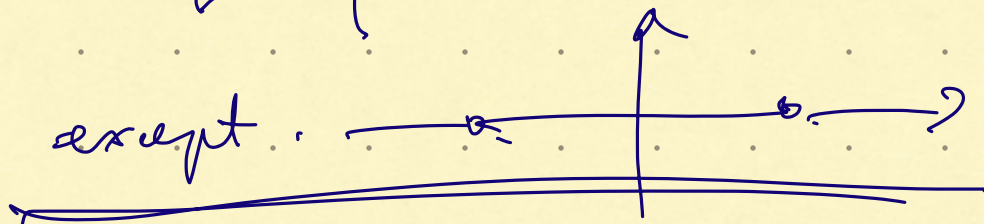
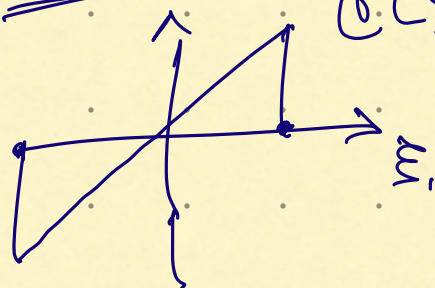
$$C(\xi) = \frac{1}{4} \sum_{m,n} A_{m,n} (1 - \xi_m \xi_n)$$

$$\max_{\xi \in [-1, 1]^N} C(\xi) = \overline{C}$$

$C(\xi)$ is poly linear.

$$\frac{\partial^2 C(\xi)}{\partial \xi_k^2} = 0$$

$$C(\xi, 1 \xi_1, \dots, 1 \xi_n) = C_0 + \sum_i \xi_i C_i$$



$$\max_{\xi \in [-1, 1]^n}$$

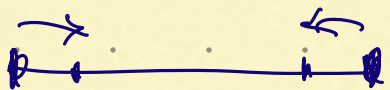
$$C(\xi) = \max_{\sigma} C(\sigma)$$

$$1) \quad \dot{\xi}_k = \frac{\partial C(\xi)}{\partial \xi_k} = - \frac{1}{2} \sum_n A_{kn} \xi_n$$

$$\xi(0). \quad C(\xi) = \frac{1}{4} \sum A_{m,n} (1 - \xi_m \xi_n)$$

$$\xi_m = \xi_m (1 - w_m) \quad \rightarrow$$

$$w_m \geq 0, \quad 0 \leq \underline{w_m} \leq 1.$$



$$C(\tilde{\xi}) = \frac{1}{4} \sum A_{m,n} [1 - \xi_m \xi_n (1 - w_m) (1 - w_n)]$$

$$C(\tilde{\xi}) - C(\xi) =$$

$$\frac{1}{2} \sum_{m,n} A_{m,n} \xi_m \xi_n w_n - \dots \frac{w_n w_m}{10^{-12}}$$

$w \sim 10^{-12}$

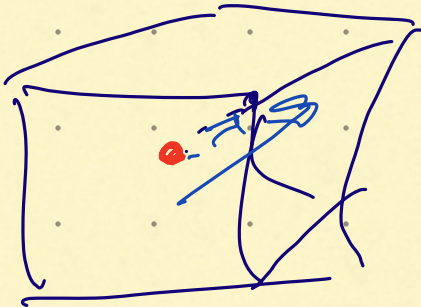
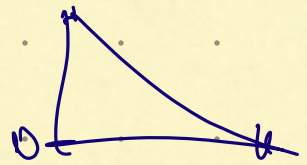
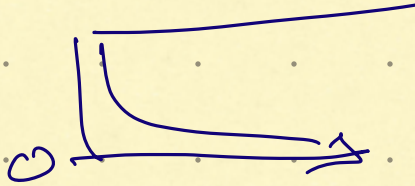
$$\frac{\partial}{\partial w_n} [C(\tilde{\xi}) - C(\xi)] = \underline{\underline{\dot{w}_n}}$$

$$\underline{\underline{\dot{w}_n}} = \frac{1}{2} \sum_{m,n} A_{m,n} \xi_m \xi_n = \underline{\underline{F_n}}$$

if $F_n < 0$, $\dot{W}_n < 0$.

and

W_n



$F_n < 0$.

$\Rightarrow F_n < 0 \Leftrightarrow \text{stable.}$

The same as LS.

But

$$\dot{\Sigma}_n = \frac{\partial C}{\partial \{x_n\}}$$

← the dyn. syst.

$$\Sigma_n(t + \Delta t) = \Sigma_n(t) + \dot{\Sigma}_n \cdot \Delta t$$

$$\Sigma(t + \Delta t) = \Phi[\Sigma(t)]$$

dynamical progression.

10^6 nodes.

$$\sigma(t + \Delta t) = \begin{cases} +1 & \text{if } \sigma(t + \Delta t) = +1 \\ & \text{is good} \end{cases}$$

$\Rightarrow$ sequential. if $\sigma(t + \Delta t) = +1$ is bad.